SMART INDIA HACKATHON 2025

TITLE PAGE

- **PROBLEM STATEMENT ID - SIH25231**
- **PROBLEM STATEMENT TITLE - Design and build a multimodal Retrieval-Augmented Generation (RAG) system leveraging a Large Language Model (LLM) for OFFLINE mode that can ingest, index, and query diverse data formats such as DOC, PDF, Images and voice recordings within a unified semantic retrieval framework**
- **THEME - SMART AUTOMATION**
- **PS CATEGORY- SOFTWARE**
- **TEAM ID - 52439**
- **TEAM NAME (Registered on Portal) - QUANTUM LEDGERS**

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IDEA TITLE

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VAULT RAG

Offline Intelligence, Unified Access

Vault RAG: Secure Multimodal Retrieval, Offline.

IDEA/SOLUTION:

Implementation of **Offline Multimodal RAG System** using LLMs that can seamlessly ingest, index, and query text, images and audio for unified semantic retrieval.

- **Offline multimodal retrieval** via unified CLIP embeddings of text, audio and images.
- **OpenAI Whisper ASR** converts local audio to searchable text without cloud.
- **FAISS Indexing** enables fast, approximate nearest neighbour search for instant results.
- Modular **FastAPI Backend** manages ingestion, embedding and real time chat-based retrieval with precise grounded citations.

OFFLINE RAG BASED PROBLEM SOLUTION

Characteristic	Unified Retrieval Source	Offline Operation & Verification	Privacy and Compliance	Emergency Response Integration
Data Source	Unified embedding space	Local audio and image processing	Context-grounded answers only	Real-time multimedia retrieval
Data Integrity	Tamper-resistant citations	Fully offline embedding and retrieval	Minimal data exposure	FAISS index traceability
Data Access	Minimizes data silos	Rapid and reliable demo performance	Consent-driven search and display	Fast citation navigation
Compliance	no	Eliminates external API dependency	Adherence to data minimization	Quick reference user interface

Semantic OpenSearch enables fast, secure offline retrieval across text, images, and audio by embedding all data into a unified vector space. It delivers explainable, context-rich results with metadata citations through an intuitive chat interface while ensuring data privacy and scalability.

To learn more about our idea via video: Video.mp4

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TECHNICAL APPROACH

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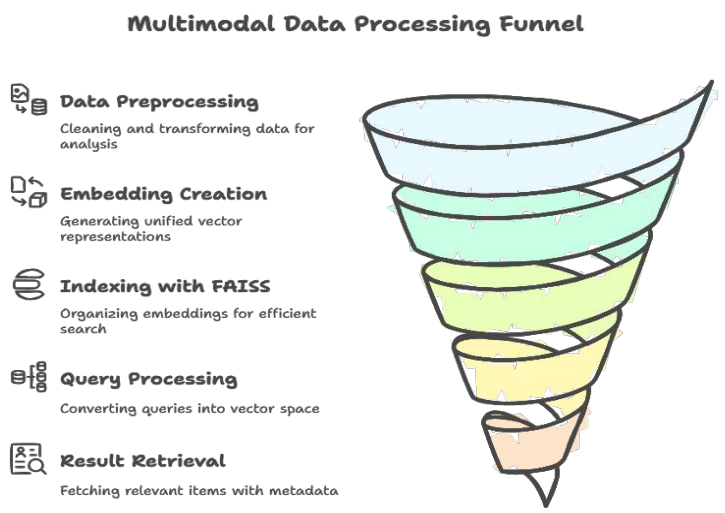
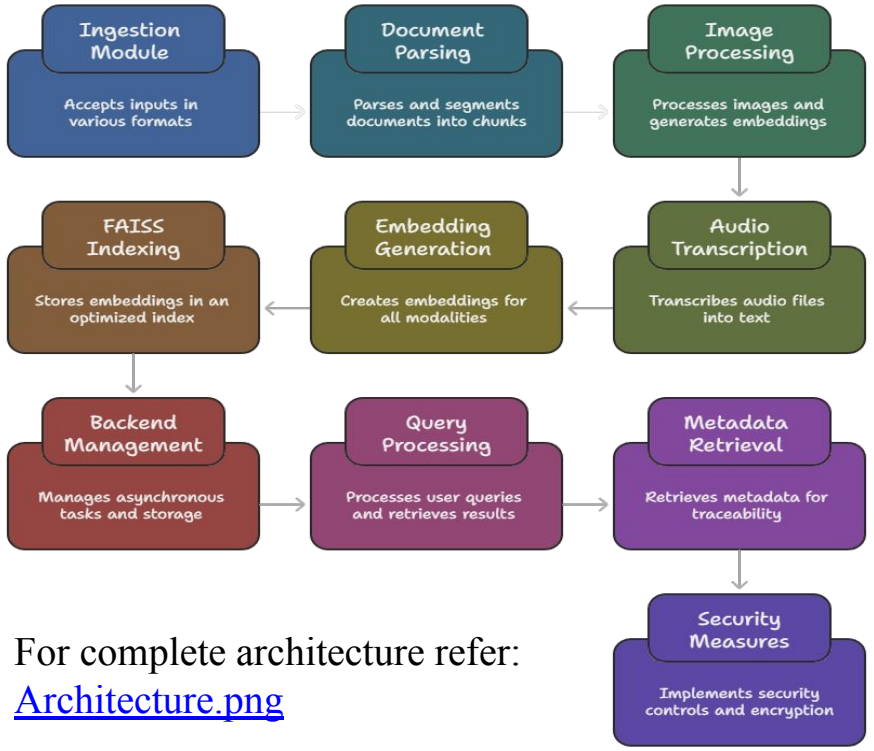
SECURITY FIRST ARCHITECTURE



- Date from text, image and audio is independently preprocessed and embedded into a **single vector space** using modality specific models.
- **FAISS indexing & Calendar based dashboard updates** enables fast offline similarity searches returning results linked with precise multi-format metadata.
- All processing stages enforce **strict security** through encrypted data storage, access controls and comprehensive audit logging to ensure **data privacy and traceability.**

IMPLEMENTATION FLOWCHART



For complete architecture refer: [Architecture.png]

TECHNOLOGY STACK

ENCODERS, Whisper, FAISS Vector Database, Python, FastAPI, Postgresql, React, Redis

IMPLEMENTATION PROTOTYPE:"Our Semantic OpenSearch prototype is a powerful, self-contained search engine that works with multiple data types, including text, images, and audio. It uses AI models to create a unified 'vector search' index that operates completely offline. A secure backend powers a chat interface where users get context-aware answers with clear citations. The system is designed for speed, security, and data confidentiality.

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FEASIBILITY AND RISK MITIGATION

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Feasibility Challenges:-

- **Diverse Data Ingestion & Normalization:** Handling varied DOCX, images, and audio formats; normalizing complex inputs for a unified semantic space.
- **Unified Semantic Embedding :** Optimizing cross-modal recall/precision, especially for domain-specific jargon, within a constrained vector space.
- **Offline Mode Performance & Scalability:** Achieving sub-second retrieval and processing speeds without dedicated GPUs; ensuring scalability for large, offline datasets.
- **Grounded Generation & Citation Accuracy:** Ensuring precise, verifiable citations and minimizing LLM hallucinations from multimodal retrieved context.

Mitigation:-

- Implement a **universal preprocessing framework** supporting DOCX, image, and audio normalization, robust OCR, and automated segmenting to streamline varied data into a unified format for further embedding and retrieval.
- **Fine-tune multimodal CLIP encoders** for domain-specific jargon, continually upgrade cross-modal recall/precision, and apply regular validation to optimize embeddings across text, images, and audio.
- Architect the **FastAPI backend** to leverage precomputed embeddings, batch processing, and optimized memory use; ensure reliable operation on standard CPUs and scale efficiently for large local datasets.
- Enforce strict **answer grounding with embedded source citations** (page, timestamp, image ID) to minimize hallucinations; develop UI cues and validation logic for verifiable, context-linked answers every time.

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IMPACT AND BENEFITS

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Impacts :-

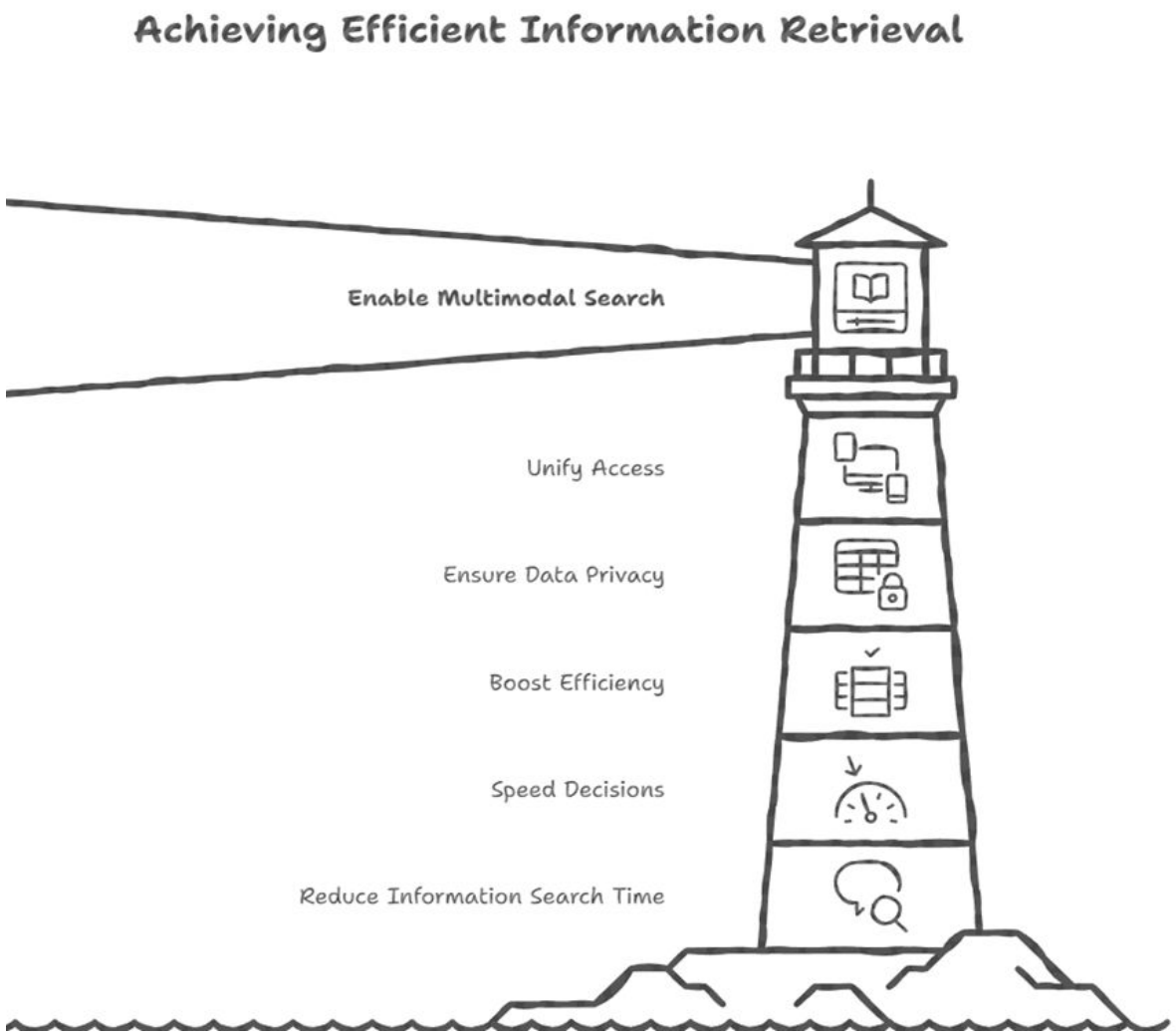
- Offline multimodal search across text, images, and audio enables fast, secure, and context-aware insights for education, healthcare, and defense.
- Cross-modal retrieval unifies access, eliminating tool-switching and boosting efficiency.

Benefits:-

- Offline encrypted processing ensures complete data privacy and compliance.
- Boosts efficiency by unifying information and reducing manual searching.

Quantifiable Analysis:-

- Speeds decisions by retrieving context-aware answers from text, images, and audio via FAISS and LLM-RAG.
- Users spend less time locating information as unified CLIP embeddings enable cross-modal search and instant retrieval.



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RESEARCH AND REFERENCES

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Technology	Description	System Usage	Reference
Input Technologies: OCR (Tesseract, Paddle OCR), Whisper AI, PDFBot-inspired parser	Advanced OCR models extract text from multilingual images and scans. Whisper AI provides accurate audio transcription. PDFBot-inspired parser extracts structured textual content including tables and lists.	Converts raw multimodal data into structured, searchable text suitable for embedding generation and downstream AI processing..	Overview of the Tesseract OCR Engine," ICDAR 2007.https://ieeexplore.ieee.org/document/43769 Radford et al., "Robust Speech Recognition via Large-Scale Weak Supervision,"https://arxiv.org/abs/2212.04356 Wang, H. et al., "Intelligent Document Parsing.", SIGIR 2025.https://doi.org/10.1145/3477495.3531930
Vector Database: (ChromeaDB)	High-performance vector database designed to store high-dimensional (512D) embeddings generated by BGE-EN-ICL, facilitating unified representation of text, images, and audio in a consistent semantic space.	Acts as the central knowledge repository for enabling efficient storage and fast retrieval of cross-modal semantic representations for AI-driven inference.	Johnson, J. et al., "FAISS: A library for efficient similarity search", arXiv 2019. https://arxiv.org/abs/1907.10565
Embeddings Model: (BGE-EN-ICL)	State-of-the-art cross-modal embedding model that generates aligned 512-dimensional vectors for diverse data modalities (text, image, audio).	Creates 512-dimensional embeddings for all data types enabling unified semantic search and retrieval, enabling user to integrate visual, auditory, and textual information seamlessly.	[Li et al. 2024] https://arxiv.org/abs/2409.15700
RAG & LLM: (Gwen 2.5-7B-Instruct + LangChain)	Large instruction-tuned large language model integrated with LangChain's RAG framework that produces context-aware multi-sourced responses.	Powers the final user-facing interface that interprets queries, synthesizes multimodal retrieved evidence, and generates precise natural language answers with source attribution.	[Yang et al. 2024] https://arxiv.org/abs/2412.15115 (Qwen 2.5 technical report)
Search Engine: (OpenSearch with kNN & FAISS)	Open-source distributed search platform supporting hybrid full-text search combined with k-Nearest Neighbor (kNN) and FAISS vector similarity search algorithms.	Enables fast and accurate retrieval of semantically similar documents and multimodal data fragments critical for autonomous robot decision-making.	[Douze et al. 2024] https://arxiv.org/pdf/2401.08281.pdf (FAISS architecture)